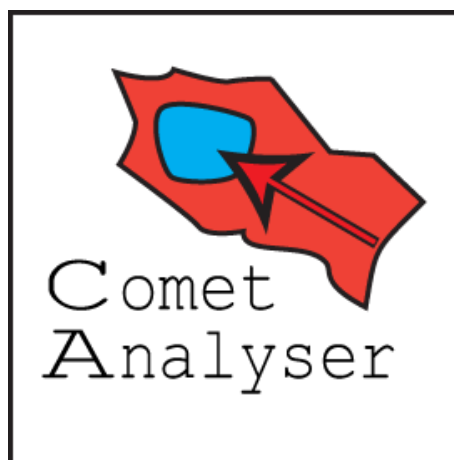


CometAnalyser – User Manual



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1. BRIEF DESCRIPTION

CometAnalyser is a user-friendly, data visualization and analysis software tool for microscopy fluorescence and silver-stained comet assay images. The main aim of CometAnalyser is to provide accurate comet segmentations and quantitative analysis using advanced image processing methods with minimal user interaction.

The comet assay provides an easy solution to evaluate the viability of cells by measuring DNA damage. This is quantified by computing the displacement by electromagnetic field of genetic material contained in the nucleus, typically called "comet head", and the genetic material in the surrounding part, considered as the "comet tail". Today, a few freely available and commercial tools exist to analyse microscopic images acquired with a comet assay. However, most of the freely available tools cannot be used for the analysis of both fluorescence and silver-stained images, and the commercial ones are costly and do not provide possibilities for examining the code and modifying the processing algorithm. To obtain reproducible and reliable quantitative data, also allowing the users to extend the number of features estimated, we developed an easy-to-use tool named CometAnalyser.

The project was started by Filippo Piccinini at IRCCS Istituto Romagnolo per lo Studio dei Tumori (IRST) "Dino Amadori", Meldola (FC), Italy, in 2017. Then, Attila Beleon in 2021 extended the tool by including machine learning modules for segmenting and classifying the comets automatically. CometAnalyser is a software routinely used in IRST IRCCS and at Biological Research Centre (BRC), Szeged, Hungary and it is provided as open-source tool to all the possible interested researchers.

This document is a short tutorial to describe the main functions of CometAnalyser. It is written for non-experts. Additional information, video tutorials, source code, and literature references are available at: <https://sourceforge.net/p/cometanalyzer>

2. LICENSE

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3. SYSTEM REQUIREMENTS

- *Source code:* CometAnalyser source code is written in MATLAB (The MathWorks, Inc., Natick, MA, USA) and works on Windows, Macintosh and Linux operating systems. It requires a MATLAB R2021a, the Image Processing Toolbox 11.2, the Statistics and Machine Learning Toolbox 12.1, the Deep Learning Toolbox 14.2, or later versions. The source code also requires the "deeplabv3plusLayers" function, that was introduced in release R2019b and is available with regular MATLAB licence at: <https://it.mathworks.com/help/vision/ref/deeplabv3pluslayers.html>
- *Standalone Version:* the standalone versions of the tool do not require a MATLAB licence, but currently (February 2022) they have been deeply tested just on Windows 64-bit systems (Windows 10, or later versions, with a CPU i7, RAM 4 GB, or higher performance computers are suggested).

4. INSTALLATION

There are two ways to use CometAnalyser:

- *Source code:* For those who have a MATLAB licence, we provide the source code. In this case, it is enough to follow these steps:
 1. Download the CometAnalyser source file from: <https://sourceforge.net/p/cometanalyzer>
 2. Extract the files from "CometAnalyser_v#.zip" to an arbitrary folder.
 3. Open MATLAB.
 4. Set the MATLAB's path to the folder containing the *CometAnalyser_GUI_App.mlapp* file.
 5. Type "CometAnalyser_GUI_App" in the MATLAB Command Window.

Upon typing “CometAnalyser_GUI_App”, the following Graphical User Interface (GUI) appears:

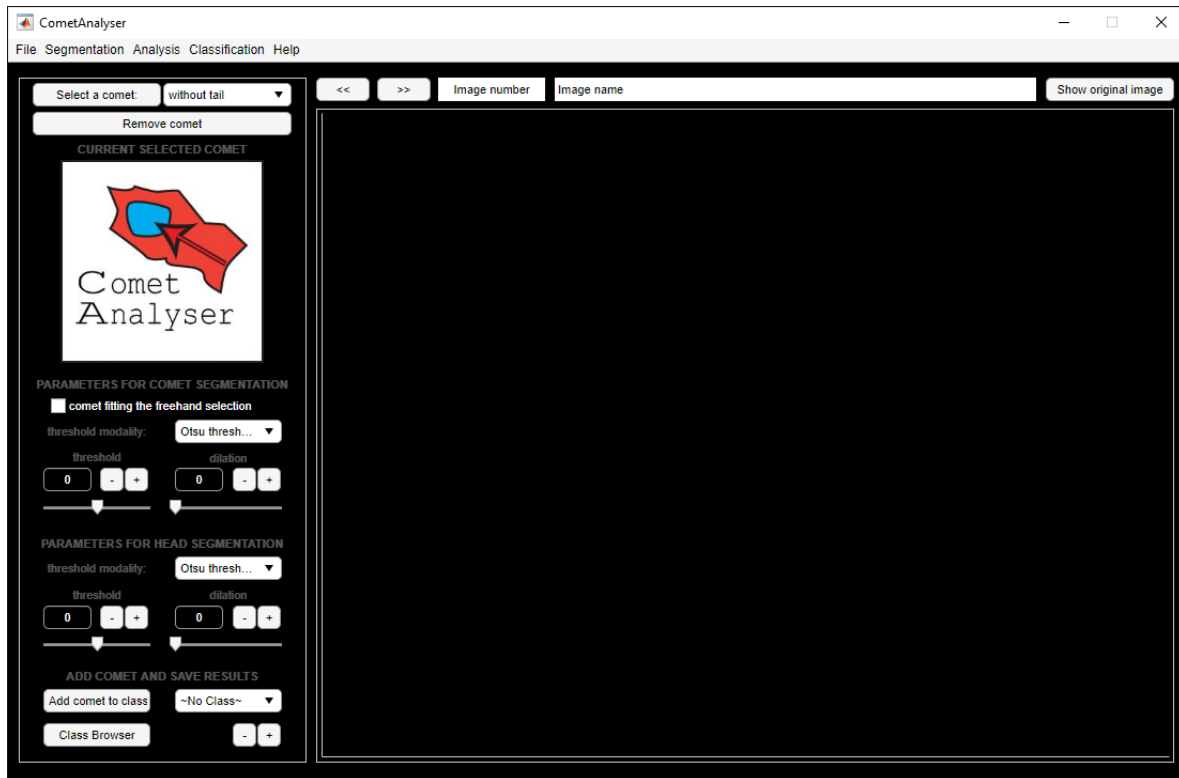


Fig. 1: CometAnalyser main window (before opening a project).

- *Win-64 Standalone Version:* To install and use the software from the Windows 64-bit standalone version you need the Administrator rights of the computer. (a) unzip the file: "CometAnalyser_v#_WIN64bit_Standalone.zip". (b) You need an Internet connection active. (c) Double-click on the file: "MyAppInstaller_web.exe" and follow the steps. (d) Once the software is installed, to run the software: Double-click on the "CometAnalyser_v#.exe" file, typically automatically generated in the "application" folder created in the selected path of the installation folder.
- *MAC and Linux Standalone Versions:* Please, carefully read the “Standalone_Linux_MAC-ARM_Win_instruction” document provided with the installation file.

At the initialization step, all the buttons have no effect until an image file is loaded. Every time a warning/error message appears, all the buttons are frozen until the message is closed.

5. INPUT IMAGES

CometAnalyser compatible with TIF, TIFF, BNP, PNG, JPG files.

Image files must have the following naming convention:

ImageName_%Num.ext

ImageName is the name of the image. The first character of *ImageName* must not be a number or a symbol. For the rest, any character can be used for the *ImageName*, except "*" and white spaces. "%Num" identifies the different individual images acquired in the same experiment. The ".ext" refers to file extension, for instance ".tif". An example of a correct file name is:

ExpA_Image_03.tif

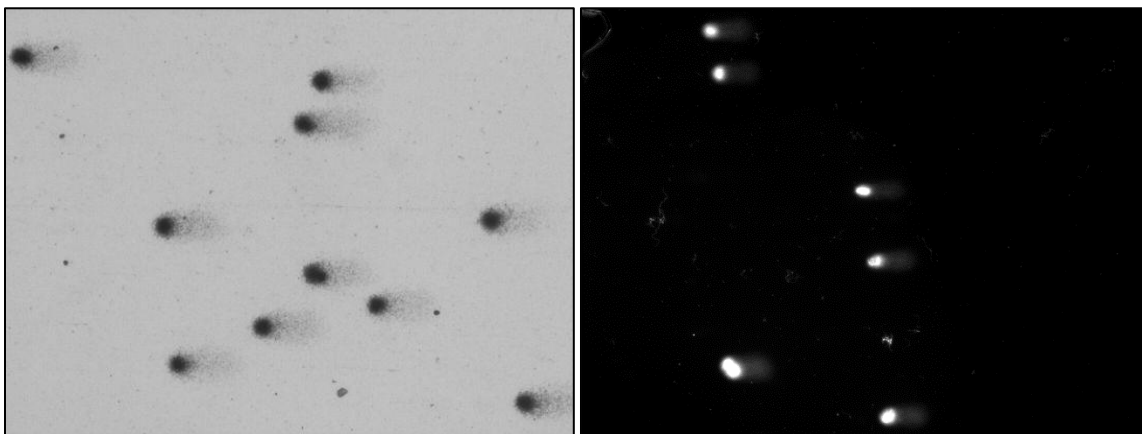


Fig. 2: Silver-stained (left) and fluorescence (right) comet assay images available at: <https://cellprofiler.org/>

A sample dataset (*Test-ProjectFolder01.zip*) to test CometAnalyser is available at the website: <https://sourceforge.net/p/cometanalyzer>

6. GETTING STARTED

As a start, follow the steps described under "**Installation**" in **Section 4.** to install CometAnalyser to your computer. Then proceed as follows:

1. Download the example dataset (named *Test-ProjectFolder01.zip*) available at the website: <https://sourceforge.net/p/cometanalyzer>
2. Extract the files into a local folder of your computer and take a look at the structure of the folder and the names of the files. File names follow the structure described in **Section 5.**
3. Open the CometAnalyser main window and click on: "File" -> "Open ..." to load the dataset. The user must select an image to be analysed, and all the files with the same extension contained in that folder will be automatically loaded into CometAnalyser. Images will be displayed with a black background and will be converted into 8-bit grayscale images. The intuitive windows reported in **Fig. 3** will appear to help the user in loading the dataset in the case of multi-channel files:

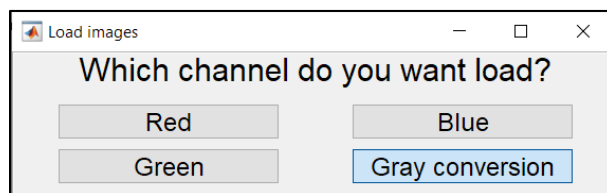


Fig. 3: GUI: load a dataset.

Generally, "Gray conversion" is the button to be used. Upon clicking on a button, the main window changes as follows:

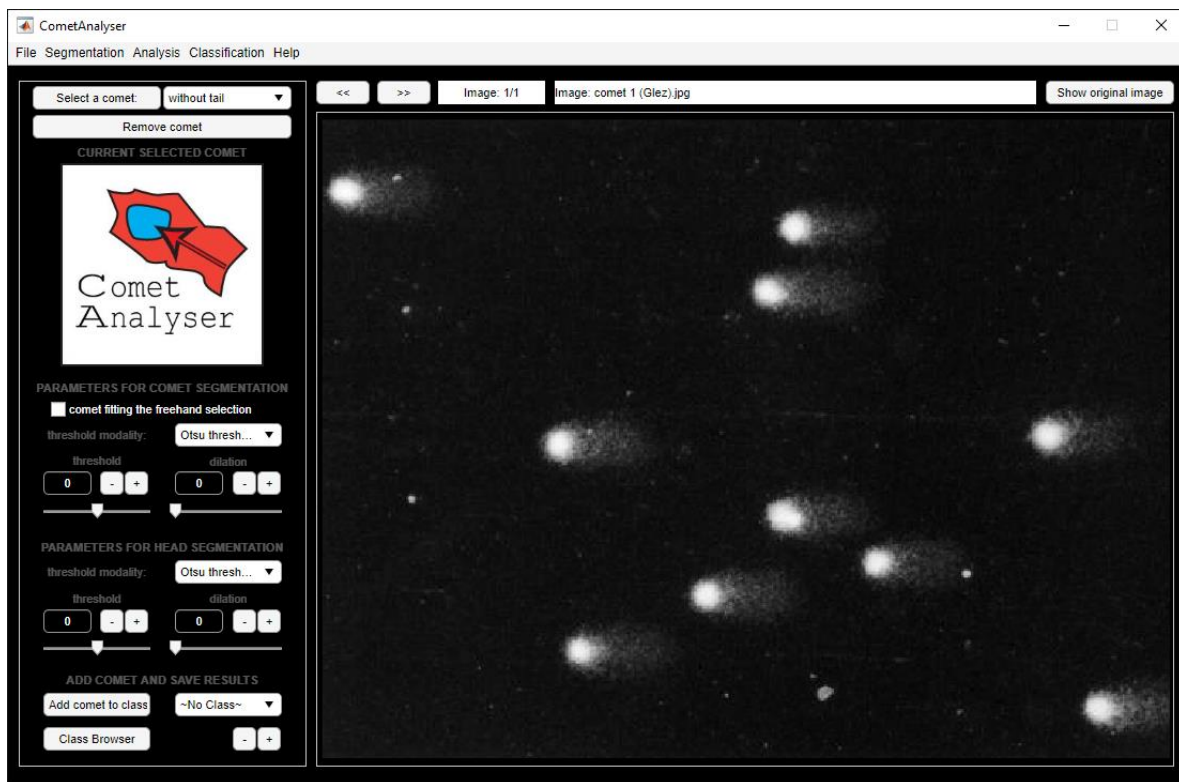


Fig. 4: CometAnalyser main window (when a dataset is loaded).

When a dataset is loaded, CometAnalyser is ready for the nucleus analysis. The main steps for any follow-up analysis include:

1. Create classes of interest.
2. Annotate cells to show different phenotypes of interest.
3. Extract quantitative features for further analysis.
4. Save the project and output the data.

All these steps, as well as the functions implemented to create classes and annotate, visualize and analyse cells, are described in detail in the following sections. However, to give a short overview, we report an example of workflow here:

Example of a typical workflow:

1. Start CometAnalyser by typing "CometAnalyser_GUI_App " in the MATLAB Command Window
2. Open a dataset with "File" -> "Open ..."
3. Create classes with "Select a comet" -> "with tail and head"/"without tail"/"without head"
4. Browse the images and annotate several cells for the defined classes by clicking on the "Select a comet" button and then selecting a cell by drawing a line around it.

5. For each comet class, adjust the segmentation parameters by trying different combinations.
6. Classify the comet with "Add comet of class" -> "Class #"
7. Export feature-based statistics to an excel sheet with "Analysis" -> "Export measurements"
8. Save data with "File" -> "Save analysed dataset"

7. GRAPHICAL USER INTERFACE (GUI)

When a new project is opened correctly (i.e. a dataset is loaded correctly), the first image of the dataset is shown in the main window. To help the user to understand the functionality of the different buttons, we proceed by describing all main sections of the GUI.

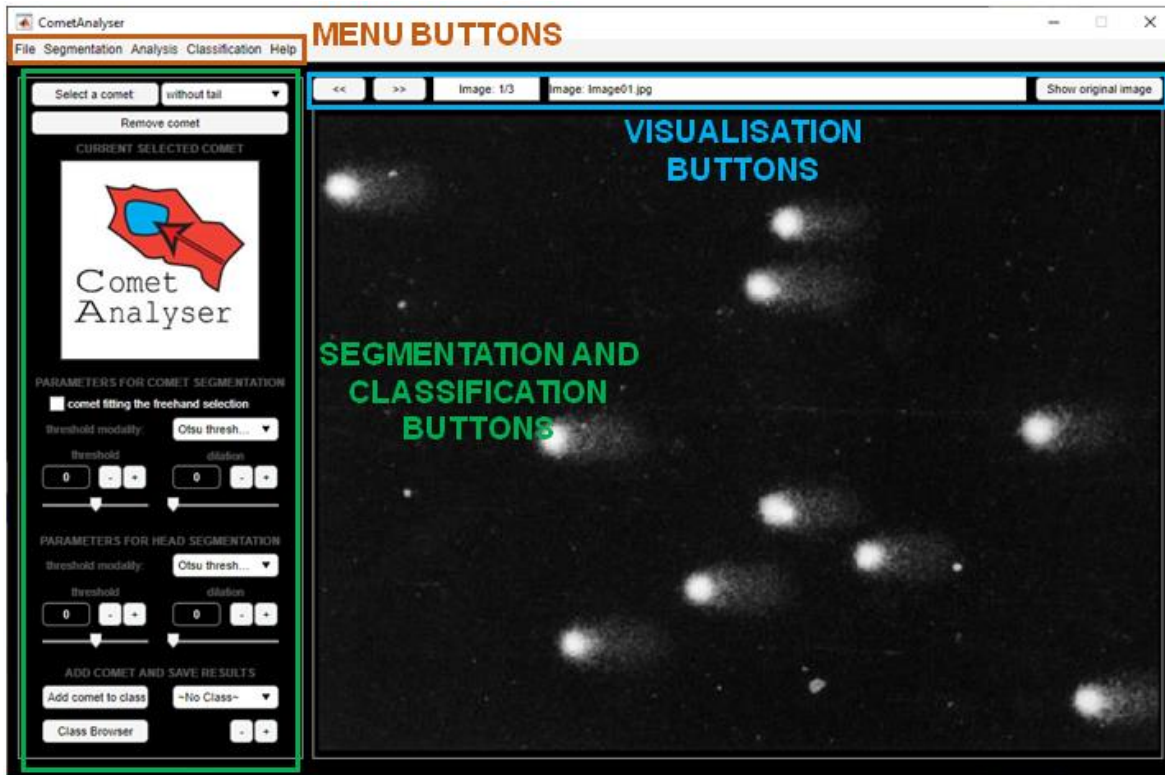


Fig. 5: GUI: main sections.

8. MENU BUTTONS

- FILE MENU

Fig. 6 shows the list of buttons included in the "File" menu.

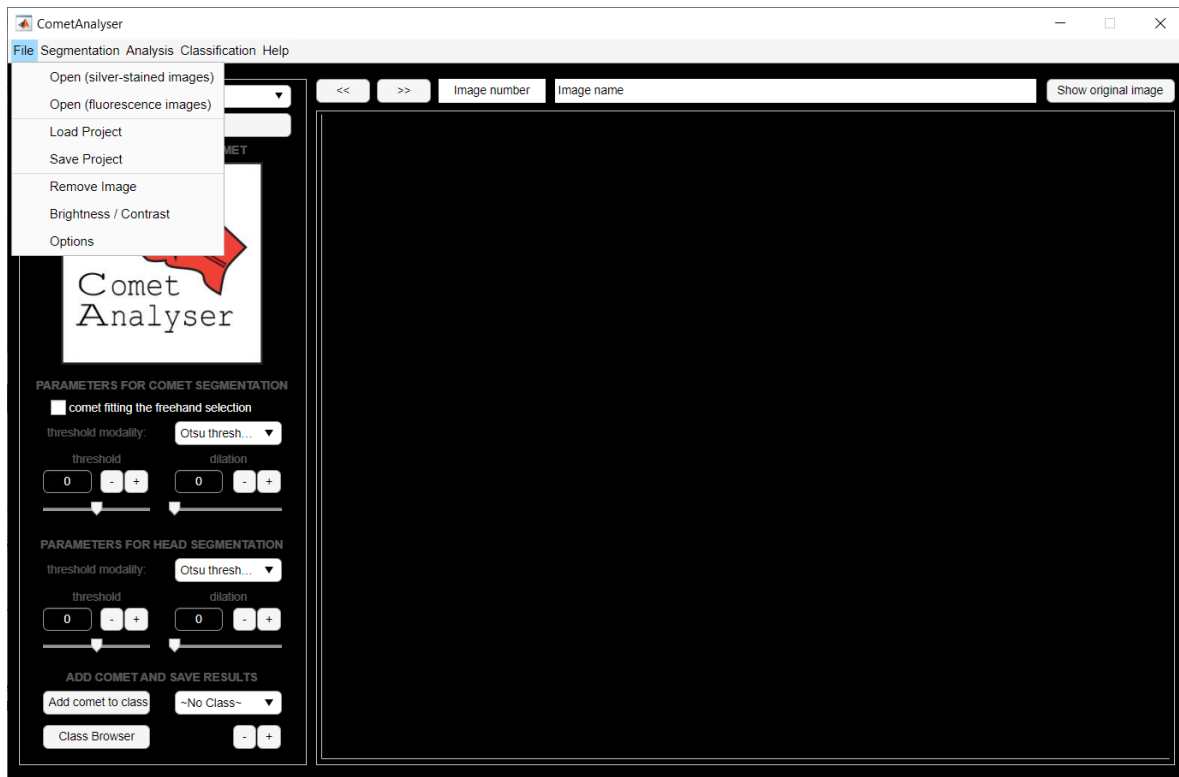


Fig. 6: File menu: list of buttons.

- "Open (silver-stained images)" and "Open (fluorescence images)" are used to load images from the "Original image folder" (see **Section 5**). All images are converted into 8-bit gray levels and are then processed to have a black background. Accordingly, silver-stained images, originally with a lighter background, are typically inverted.
- "Save Project" and "Load Project" are used to save and load back analysed images, with all the comet segmented and annotated. The "Save project" function is also automatically called every time that 10 new comets have been manually classified.
- "Remove Image" allows to remove the currently visualised images and all the related annotations.

- “Brightness / Contrast” allow pre-processing the loaded images for better visualisation of the comets’ tail.
- “Options” opens a wizard to set some parameters used in the project management (e.g. the auto-saving option for the projects).

- SEGMENTATION MENU

Fig. 7 shows the list of buttons included in the "Segmentation" menu.

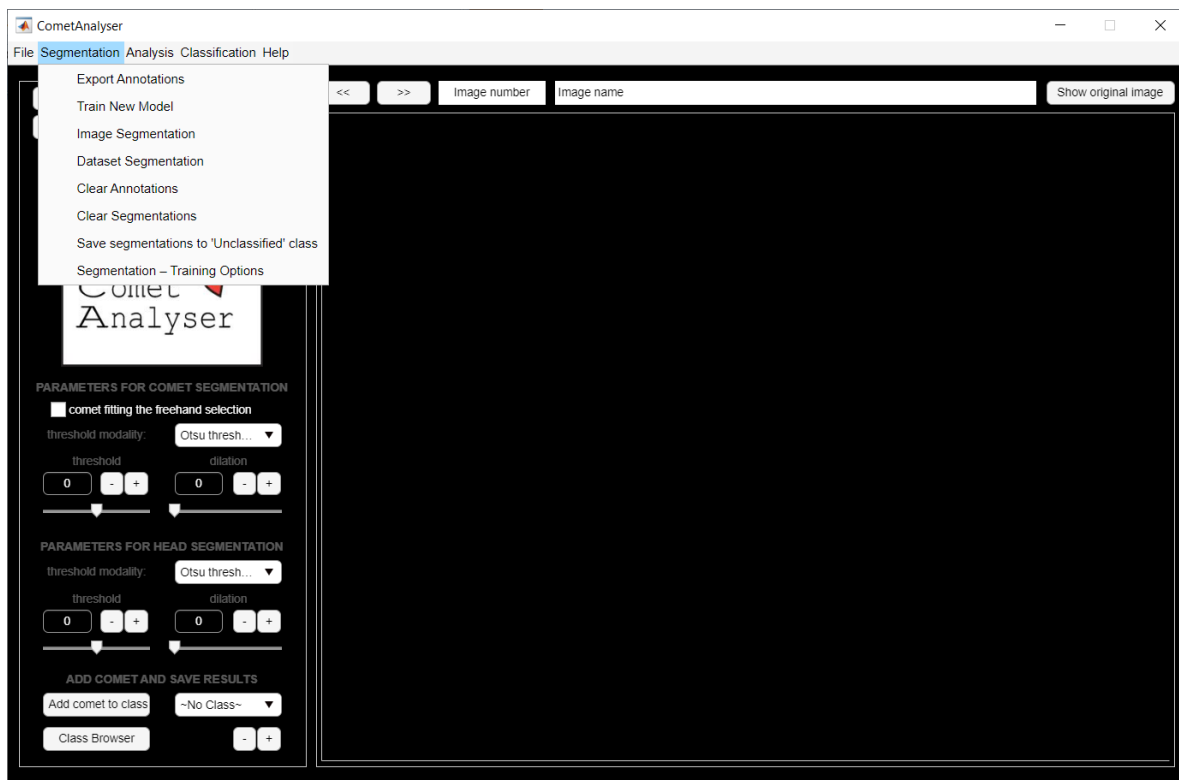


Fig. 7: Segmentation menu: list of buttons.

- "Export Annotations" is used to compute and export the annotations done. Precisely, two subfolders are created upon clicking on the "Export Annotations" button, one containing the original annotated images and one containing the masks created.
- “Train New Model” allows creating a new model for recognising new comets in the images and automatically annotating them accordingly with the comets previously annotated. Upon clicking on

the “Train New Model” button, a pop-up window leads the user in browsing the folders of the computer to select the one containing the “Exported Annotations”. The trained model is then used for automatically segmenting the comets using a convolutional neural network 18 layers deep, known with the name “ResNet-18”:
<https://www.mathworks.com/help/deeplearning/ref/resnet18.html>

- “Image Segmentation” is used to segment the comets in a specific image according to a model previously created using the “ResNet-18” convolutional neural network. Similarly, “Dataset Segmentation” is used to segment the comets in all the non-previously annotated images. Please, note that a pre-trained segmentation model logically can perform well just on images with characteristics similar to those of the images used for training the models. However, the user can easily train a new network following these steps:
 - (a) Annotate an appropriate number of comets. Please note that all the comets in the images included in the training set must be segmented (i.e. images in the training set cannot be just partially segmented).
 - (b) Export the segmentation masks using the “Segmentation - Export Annotation” button of CometAnalyser.
 - (c) Train a new deep-learning model using the “Segmentation - Train New Model” button of CometAnalyser. Network parameters can be modified using the "Segmentation – Training Options" button.
- “Clear Annotations” allows to clean all the annotations present in the visualised image.
- “Clear Segmentation” allows to clean all the segmentations automatically created using the buttons “Image Segmentation” and “Dataset Segmentation”.
- “Save Segmentations to “Unclassified” class” saves all the segmented comets not already saved in a specific class, to a new class automatically created with the name “Unclassified”. If the “Unclassified” class is already present, the new comets are included in it. This class is visible in the Class Browser and the comets in the “Unclassified” class can be manually moved in the classes created by the user to be then useful for training a classifier. This is also the class used to define the comets to be then classified even automatically using the button “Classification” -> “Prediction/Classification”.

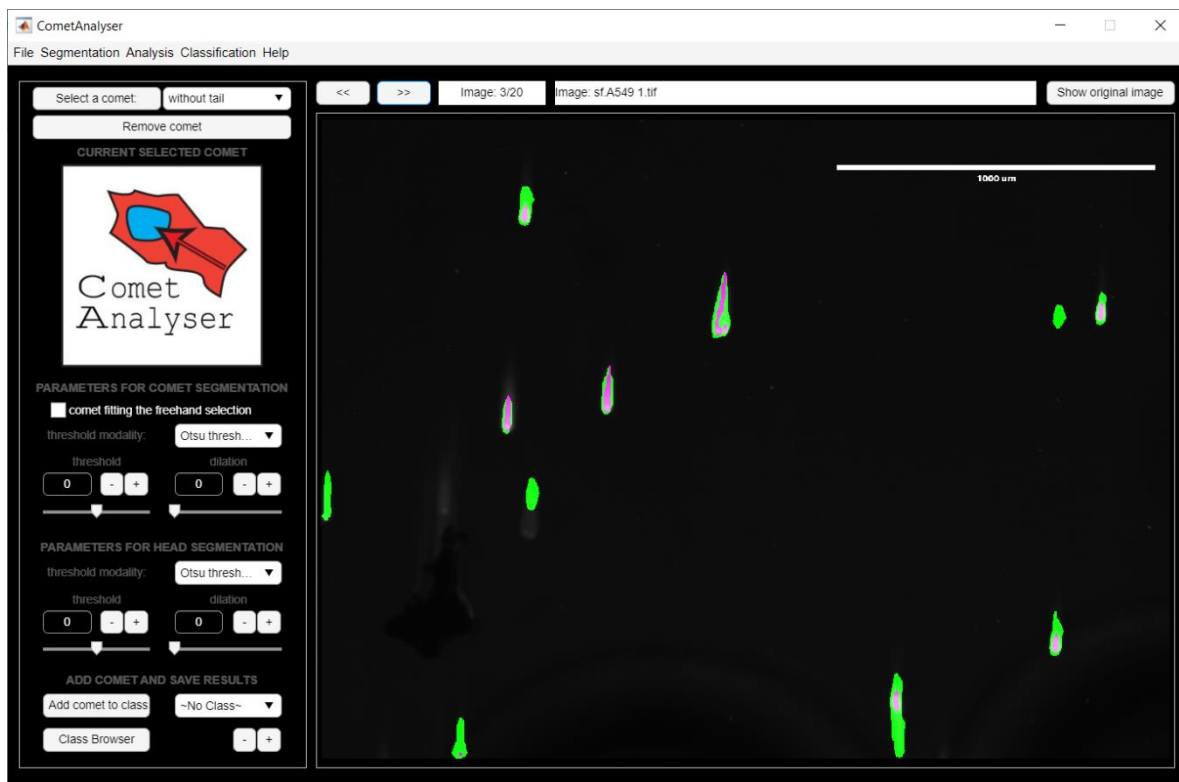


Fig. 8: Automatically segmented comets.

CometAnalyzer uses a different colour scheme (green and pink) to highlight automatically segmented comets.

- “Segmentation – Training Options”. A new window opens to allow the user to modify several parameters then considered during the training of a new model. The full list of options available for the deep learning model implemented (i.e. “ResNet-18”) is reported at: https://it.mathworks.com/help/deeplearning/ref/trainingoptions.html#mw_4f311de8-ad53-4201-ac1a-47f66e0de52d. However, let us report here the description of these parameters:

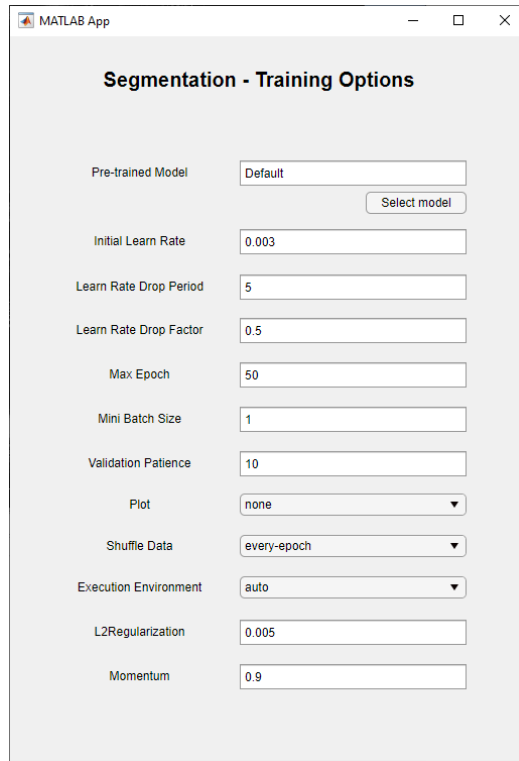


Fig. 9: Training options GUI.

- (a) Pre-trained Model: in case of availability of a pre-trained model, just search for it using the “select model” button.
- (b) Initial Learn Rate: Initial learning rate used for training, specified as a positive integer (the default value is 0.003). If the learning rate is too low, then training takes a long time. If the learning rate is too high, then training might reach a suboptimal result or diverge.
- (c) Learn Rate Drop Period: Number of epochs for dropping the learning rate, specified as a positive integer. The software multiplies the global learning rate with the drop factor every time the specified number of epochs passes-
- (d) Learn Rate Drop Factor: Factor for dropping the learning rate, specified as a scalar from 0 to 1. LearnRateDropFactor is a multiplicative factor to apply to the learning rate every time a certain number of epochs passes.
- (e) Max Epoch: Maximum number of epochs to use for training, specified as a positive integer. An iteration is one step taken in the gradient descent algorithm towards minimizing the loss function using a mini-batch. An epoch is the full pass of the training algorithm over the entire training set.

- (f) **Min Batch Size:** Size of the mini-batch to use for each training iteration, specified as a positive integer. A mini-batch is a subset of the training set that is used to evaluate the gradient of the loss function and update the weights.
- (g) **Validation Patience:** Patience of validation stopping of network training, specified as a positive integer or Inf. ValidationPatience specifies the number of times that the loss on the validation set can be larger than or equal to the previously smallest loss before network training stops. If ValidationPatience is Inf, then the values of the validation loss do not cause training to stop early.
- (h) **Plot:** to display during network training the training progress. In particular, the plot shows mini-batch loss and accuracy, validation loss and accuracy, and additional information on the training progress. The plot has a stop button in the top-right corner. Click the button to stop training and return the current state of the network.
- (i) **Shuffle Data:** 'once' — Shuffle the training and validation data once before training. 'never' — Do not shuffle the data. 'every-epoch' — Shuffle the training data before each training epoch, and shuffle the validation data before each network validation. If the mini-batch size does not evenly divide the number of training samples, then the training discards the training data that does not fit into the final complete mini-batch of each epoch. To avoid discarding the same data every epoch, set the 'Shuffle' value to 'every-epoch'.
- (j) **Execution Environment:** Hardware resource for training network, specified as one of the following: 'auto' — Use a GPU if one is available. Otherwise, use the CPU. 'cpu' — Use the CPU. 'gpu' — Use the GPU. 'multi-gpu' — Use multiple GPUs on one machine, using a local parallel pool based on your default cluster profile. If there is no current parallel pool, the software starts a parallel pool with a pool size equal to the number of available GPUs.
- (k) **L2Regularization:** Factor for L2 regularization (weight decay), specified as a nonnegative scalar.
- (l) **Momentum:** Contribution of the parameter update step of the previous iteration to the current iteration of stochastic gradient descent with momentum, specified as a scalar from 0 to 1. A value of 0 means no contribution from the previous step, whereas a value of 1 means maximal contribution from the previous step.

- ANALYSIS MENU

Fig. 10 shows the list of buttons included in the "Analysis" menu.

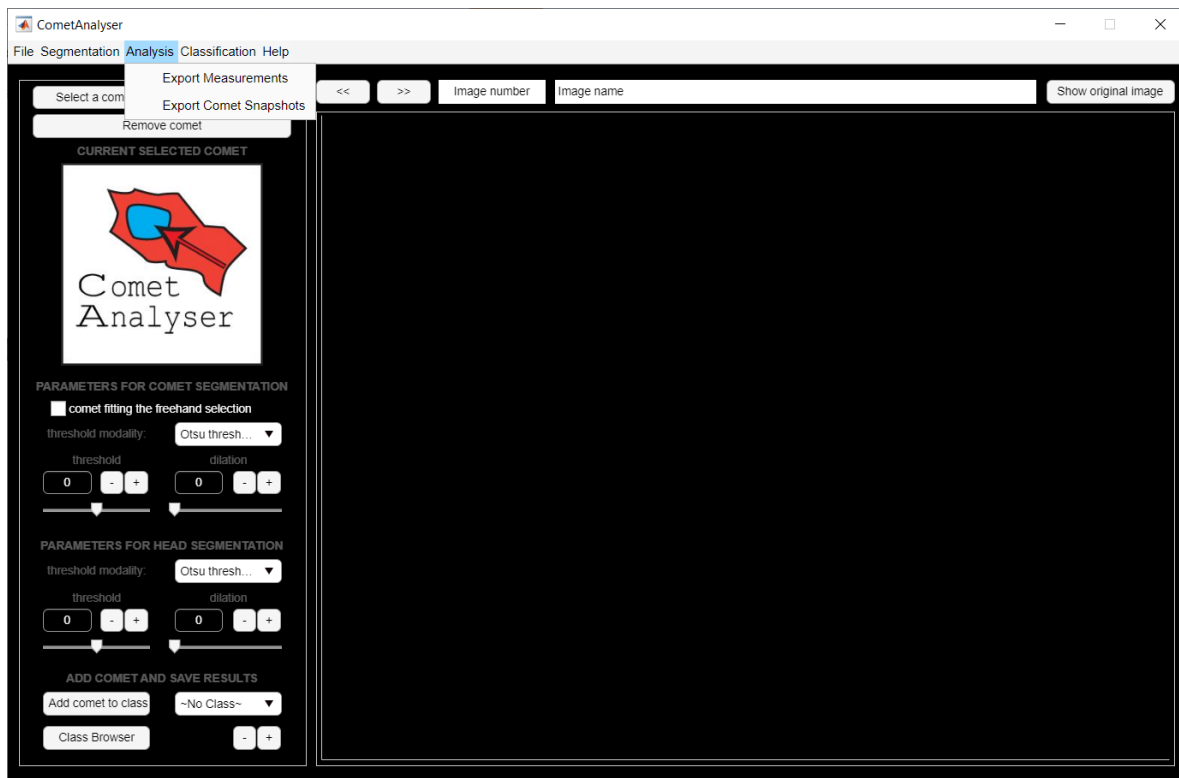


Fig. 10: Analysis menu: list of buttons.

- "Export Measurements" is used to compute and export as an Excel table (it is saved as a ".csv" file) several intensity and morphological features for each annotated comet. This is the file required when asking for training a new classification model (see: "Classification" -> "Train New Model").
- "Export Comet Snapshot" allows exporting a small snapshot for each classified comet, subdividing the snapshot according to the different classes defined in the Comet Browser.

- CLASSIFICATION MENU

Fig. 11 shows the list of buttons included in the "Classification" menu.

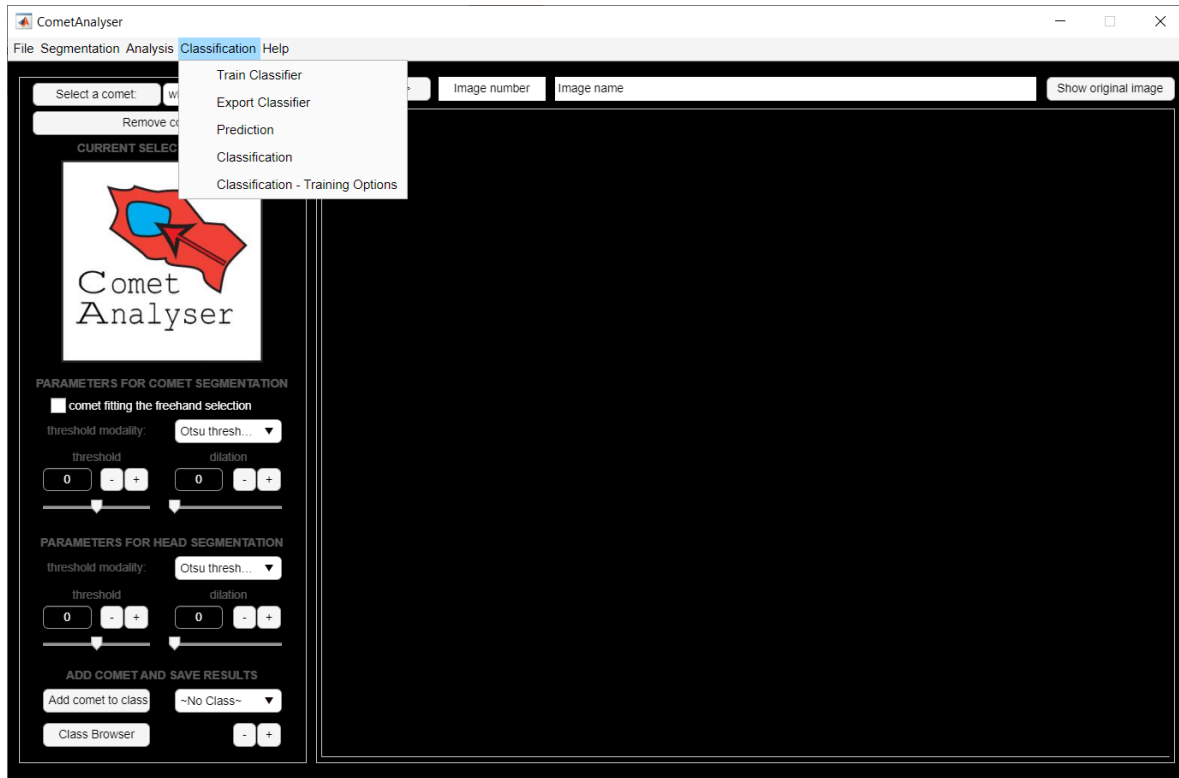


Fig. 11: Classification menu: list of buttons.

- “Train Classifier”: To compute a new classifier according to the already classified comets. To train a new classifier, the user must:
 - (a) define the classes and manually classify an appropriate number of comets for each class;
 - (b) Export the features (i.e. “Analysis – Export Measurements”) of the classified comets;
 - (c) Define the machine learning algorithm to be used for creating the network.
- “Export Classifier”: To save the classifier for re-use later or within similar projects.
- “Prediction”: To visualise for each segmented comet, the currently predicted class.
- “Classification”: To save in pre-defined classes the currently unclassified comets, according to the prediction values.

- “Classification – Training Options”: A new window opens to allow the user to modify several parameters then considered during the training of a new model. However, let us report here the description of these parameters:

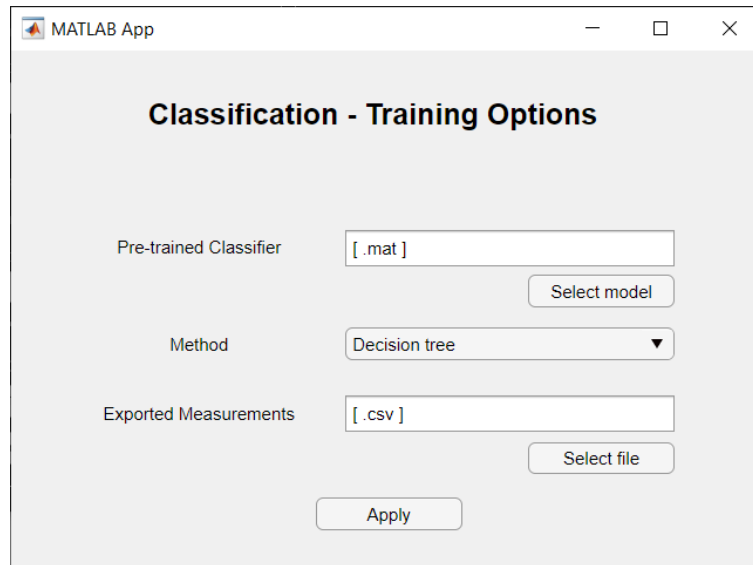


Fig. 12: Classification Training Options GUI

- “Pre-trained Classifier”: in case of availability of a pre-trained classifier, just search for it using the “select model” button.
- “Method”: We provided several machine learning algorithms to define the classification network, for instance: “Decision Tree”, “k Nearest Neighbour”, “Naive Bayes”, “Support Vector Machine”.
- “Exported Measurements”: The exported measurements are fundamental for two different procedures. (a) They are required if the user wants to train a new classification model. In this case, he/she must export metadata (i.e. features) of pre-classified comets to allow the machine learning algorithm to define the new classification network. If a pre-created metadata file exists, just search for it using the “select file” button. (b) Metadata are also required when the user wants to automatically predict/classify a comet using a pre-trained model. In this case, CometAnalyser needs metadata (i.e. the features) of the comets to be classified. To do that, first, the user must save the segmentations of the comets to be predicted/classified to the “Unclassified” class. Then, he/she must export the metadata using the “Analysis – Export Measurements” button.

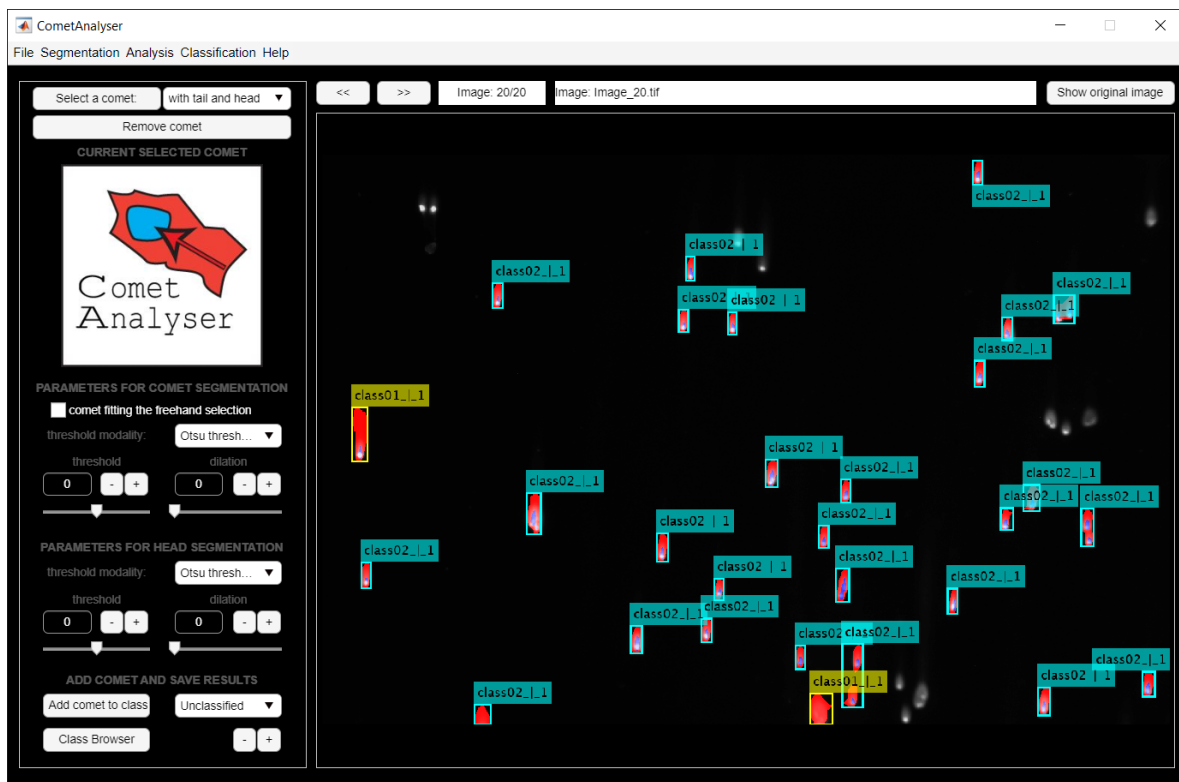


Fig. 13: Class prediction for the comets saved into the “Unclassified” class. Next to the class name there is the confidence value of the prediction.

- HELP MENU

Fig. 14 shows the list of buttons included in the "Help" menu.

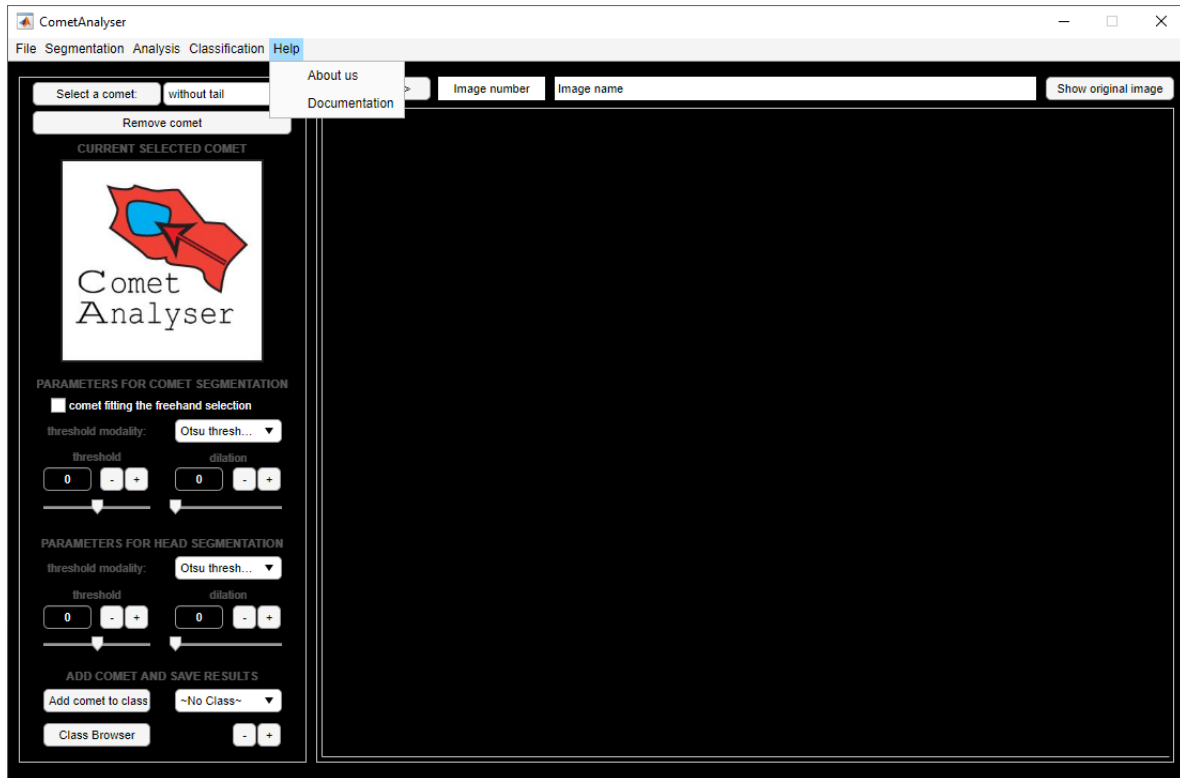


Fig. 14: Help menu: list of buttons.

The "Help" menu just shows information about software distribution, contacts and documentation in general.

9. SEGMENTATION AND CLASSIFICATION

Fig. 15 shows the left side of the main GUI of CometAnalyser, with the available options for manual segmentation and classification.

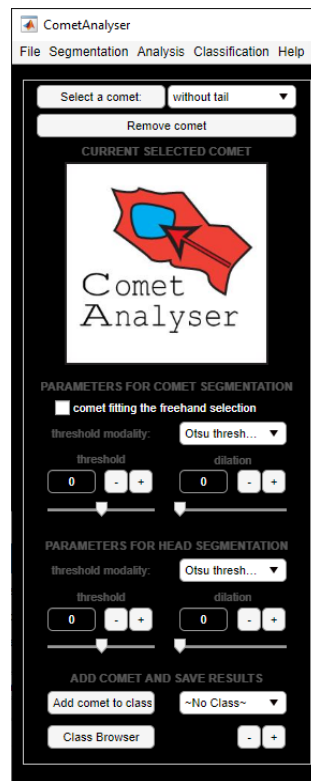


Fig. 15: Segmentation and classification buttons.

1. First of all, the user must define if the comet is "with tail and head"/"without tail"/"without head".
2. After clicking on the "Select a comet" button, the user must draw a ROI (typically a circle) in the main window around the comet of interest.
3. Once he will press double-click on the ROI defined, the comet will be displayed into the "Current selected comet" window (on the left side of the GUI of CometAnalyser), as shown in **Fig. 16**.

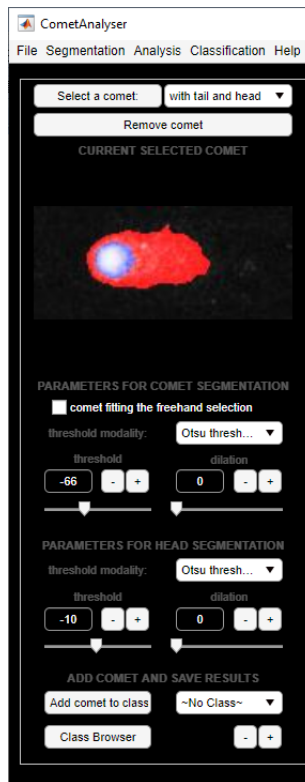


Fig. 16: Segmented comet preview.

4. We suggest trying the different “threshold modalities” available, to see which one better fits the selected comet. Currently, three different threshold modalities are available. All of them are based on the analysis of the grey-level intensity histogram of the local ROI including the comet. The first is the classical Otsu thresholding segmentation [Otsu, N. “A threshold selection method from gray-level histograms”, IEEE Trans. Syst. Man Cybern. 9 (1) (1979) 62–66.]. The second is the triangle segmentation [Zack, G.W. *et al.*, “Automatic measurement of sister chromatid exchange frequency”, J. Histochem. Cytochem. 25 (7) (1977) 741–753.]. Finally, we implemented an additional segmentation strategy called “Average between Otsu and Triangle”, with the final threshold defined as the average value of the ones defined according to the two methods just cited.
5. Several buttons are available to modify the threshold and the dilation of the masks of the “comet head” and “comet tail”.
6. Once the user is satisfied with the segmentation obtained, by pressing the “Add comet to class” button the current segmentation will be saved to an arbitrary class (classes must be pre-defined using the “–” and “+” buttons on the bottom part of the left side of the GUI of CometAnalyser).

Once manually classified, the segmented comet will be displayed with colours also in the main window.

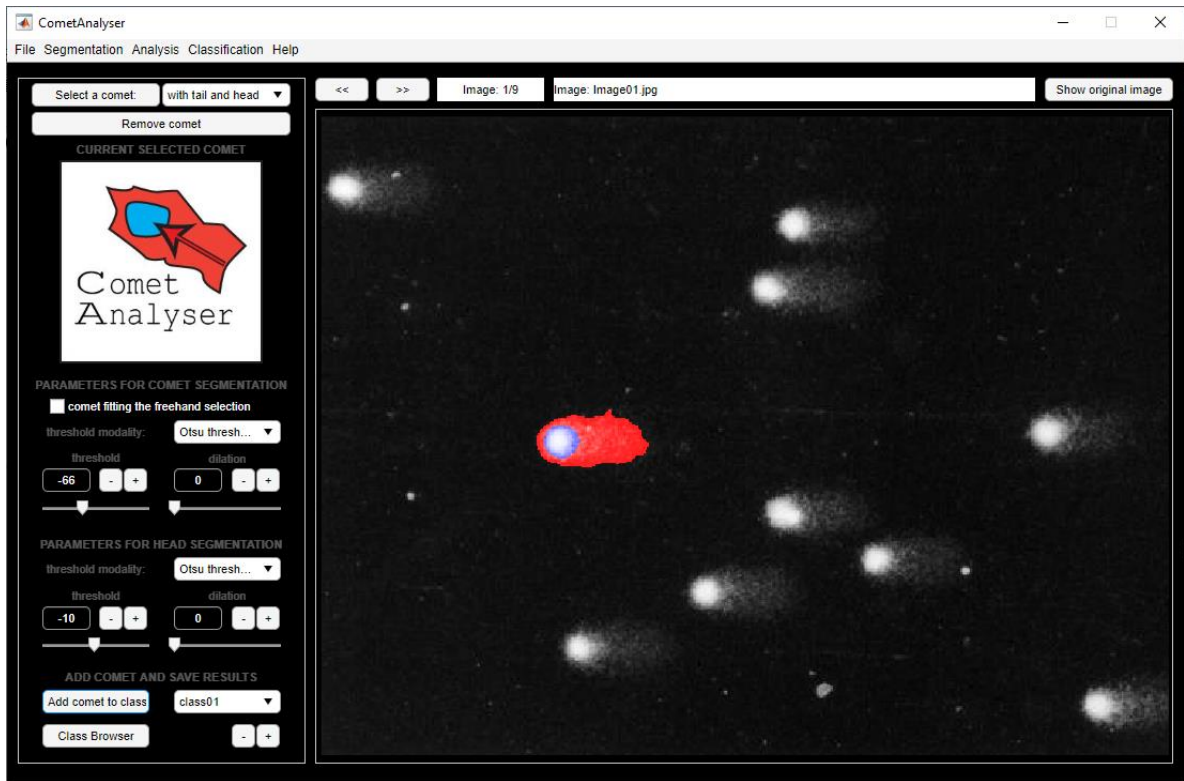


Fig. 17: Segmented comet shown in the main window.

To remove a pre-segmented comet:

1. Select a comet by clicking on it, or using the previously explained method.
2. The comet will be visualized in the "Current selected comet" window. Just click on the "Remove comet" button.

To better understand the segmentation output, we show the mask for the head and the mask for the tail for a segmented comet, showing that the mask of the tail + the mask of the head are not composing the full mask of the comet for the border around the head. The 100% of the "Comet" mask, for us, is computed using: "Head" mask + "Comet-Head" mask, and not using: "Head" mask + "Tail" mask. Now that our definitions of the different masks are clear, it is easier to understand why the sum of Head DNA and Tail DNA, with these definitions should not be 100%.

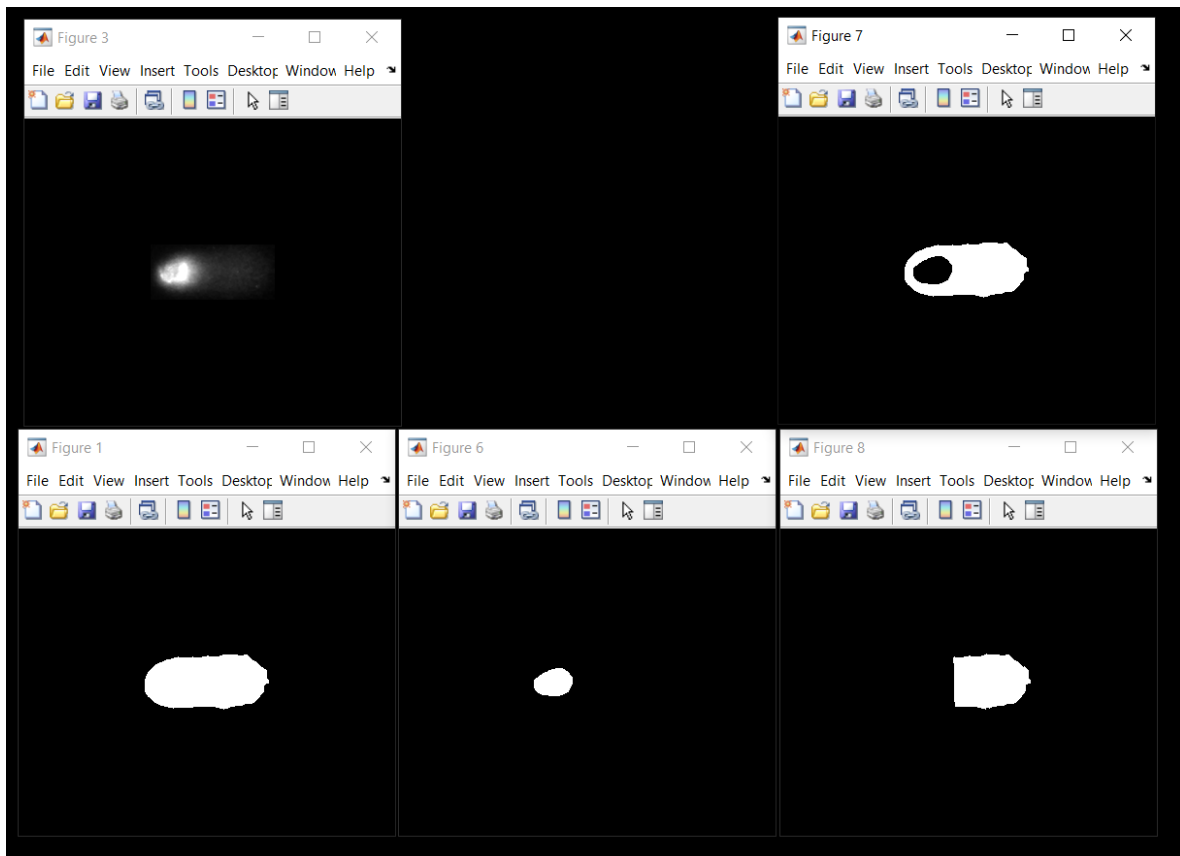


Fig. 18: Definitions of the different masks: (Top-Left) original Comet; (Bottom-Left) "Comet" mask; (Bottom-Central) "Head" mask; (Bottom-Right) "Tail" mask; (Top-Right) "Comet-Head" mask.

- AUTOMATIC SEGMENTATION

As already anticipated, CometAnalyser provides a deep learning method based on a convolutional neural network 18 layers deep, known with the name "ResNet-18" (<https://www.mathworks.com/help/deeplearning/ref/resnet18.html>), to automatically segment the comets in a specific image or in the full dataset.

The user can:

1. Exploit a pre-trained segmentation model. A provided pre-trained segmentation models logically can perform well just on images with characteristics similar to those of the images used for training the models.

2. Train a new segmentation model by exporting previously segmented annotations and then use it to segment the comets in images not containing annotations.

If a pre-trained segmentation model is available, the user simply has to select it by clicking on “Segmentation” -> “Segmentation – Training Options” to define the path of the model (previously saved as a “.mat” file). Otherwise, if the user wants to train a new segmentation model, first he/she must save the annotations using the button “Segmentation” -> “Export Annotations” (two subfolders will be automatically created, one called “Images” and one called “Masks”, containing respectively the original annotated images and the masks created) and then he/she has to click on “Segmentation” -> “Train New Model”. Note that for training a new model several options are available at “Segmentation” -> “Segmentation – Training Options”.

- AUTOMATIC CLASSIFICATION

A single comet or multiple comets can be automatically classified using a pre-trained classification model or a new one trained using specific annotations.

If a pre-trained classification model is available, the user simply has to select it by clicking on “Classification” -> “Classification – Training Options” and defining the path of the model (previously saved as a “.mat” file). Otherwise, if the user wants to train a new classification model, first he/she must export the measurements of previously classified comets using the “Analysis” -> “Export Measurements” button (a “.csv” file will be automatically created) and then he/she has to click on “Classification” -> “Train Classifier”. For defining the classifier architecture, several options are available at “Classification” -> “Classification – Training Options”.

Once defined a classifier, the user must export as a “.csv” file the measurements of all the comets, including those in the “Unclassified” class using the “Analysis” -> “Export Measurements” button. Then, he/she has to set the path of the created “.csv” file using the “Classification” -> “Classification – Training Options” button. Finally, when clicking on “Classification” -> “Prediction”, the confidence value and class of all the comets in the “Unclassified” class of the Comet Browser will be displayed using coloured labels in the main window of CometAnalyser. To confirm the prediction for all the comets, the user can click on “Classification” -> “Classification”. Otherwise, he/she can click on a single comet and manually accept the suggested class by selecting the class on the list on the left side of the GUI and pressing the “Add comet to class” button.

10. CLASSES

Please, not the “Class Browser” button on the bottom-left side of the GUI of CometAnalyser, reported in **Fig. 18**.

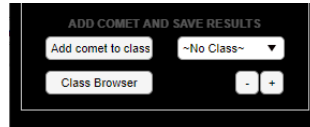


Fig. 19: Class Browser opening button.

The “Class Browser” button opens the GUI shown in **Fig. 19**:

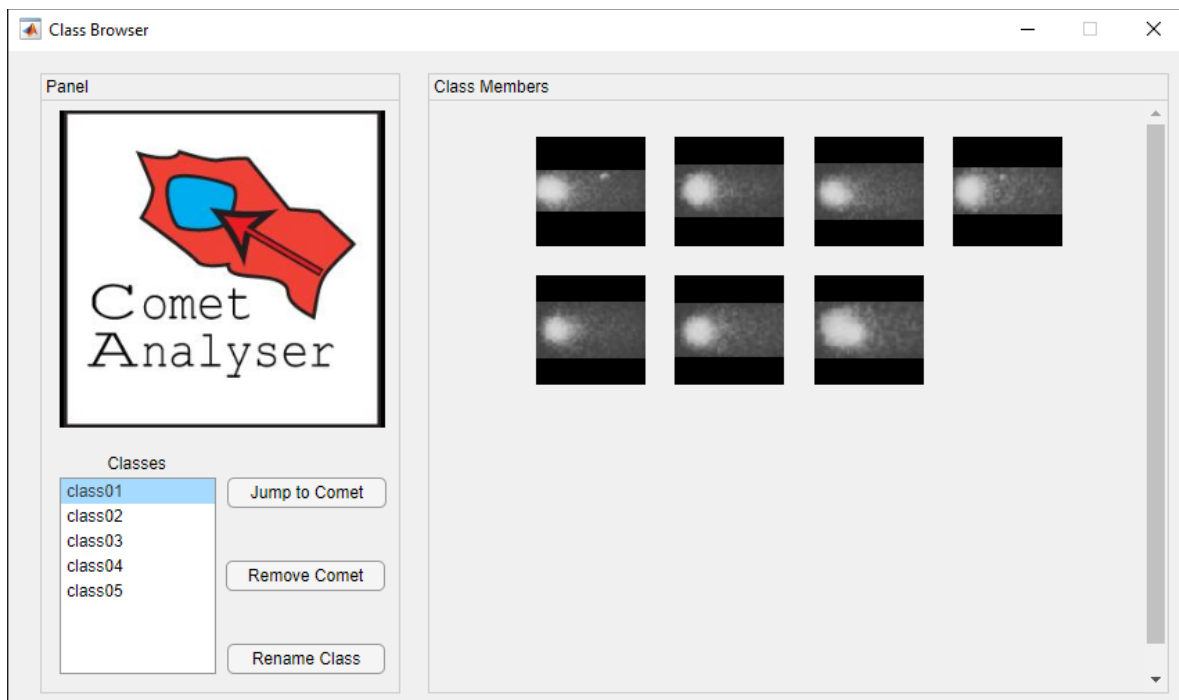


Fig. 20: Class Browser GUI.

- “Panel” shows the currently selected comet on the window named “Class Members” on the right side of the GUI.
- “Class Members” lists all the comets classified for the selected class in the “Classes” panel.

- “Classes” lists the names of all the classes created with the “+” button on the bottom-left part of the main GUI of CometAnalyser. Currently, there is no limit for the maximum number of created classes.
- “Jump to Comet” opens the image of the currently selected comet in the main window of the main GUI of CometAnalyser.
- “Remove Comet” allows to remove from the list and in general from the annotations the comet currently selected in the Class Browser.
- “Rename Class” allows renaming an existing class.

11. VISUALISATION BUTTONS

Fig. 20 shows the buttons available in CometAnalyser to change the image displayed in the main window.



Fig. 21: Visualisation buttons.

The first two buttons (i.e., “<<” and “>>”) allow scrolling the images loaded. The first text-field reports the total number of loaded images and the number of the displayed image. The second text-field reports the name of the displayed image. Finally, the “Show original image” button allows displaying the original image, without the 8-bit, gray level conversion automatically applied by CometAnalyser.

12. OUTPUT

Using the “File - Save Project” button the user can save the current project with all the defined classes and segmented comets as a “.mat” file. This file can be then easily reopened for further analysis using the “File - Load Project” button.

Using the “Segmentation - Export Annotations” button the user can export as gray-level “.png” images the masks of the segmented comets. Pixels labelled with 0 belong to the background, with 127 belonging to the tails of the comets, with 255 belonging to the heads of the comets.

Using the “Analysis - Export Comet Snapshots” button the user can export as gray-level “.tiff” images the snapshots of the segmented comets subdivided into sub-folders with the names of the classes reported in the “Class Browser” tool.

Using the “Analysis - Export Measurements” button the user can export as a comma-separated value file, (a “.csv” file) readable with the classical Excel/Calc software the intensity and morphological features automatically computed for each segmented comet.

In addition, we provided an easy solution for users interested in extending the number of features by implementing new ones.

To implement a new feature the user has to open the MATLAB source code and run it following these steps:

- 1) Open the “CometAnalyser_SourceCode_v#.#” folder containing the MATLAB source code of CometAnalyser and then the “DATAfunctionsRepository” sub-folder;
- 2) Copy the “data_TemplateFunction.m” file from the “DATAfunctionsRepository” sub-folder to the “DATAtoBeExtracted” sub-folder;
- 3) Change the name of the “data_TemplateFunction.m” file to the name of the feature to be implemented, for instance change the name of the “data_TemplateFunction.m” file to “data_FeatureName01.m”). Note that the first 5 characters in the file’s name must remain “data_” and no white spaces and symbols can be used;
- 4) Open the “data_FeatureName01.m” with MATLAB. A code similar to the following one will be visualised:

```
function OutValue = data_TemplateFunction(Maski_Origj, Maski_Cometj, Maski_Tailj,
Maski_Headj, Intensity_MinMax_StretchedOrig)
% To extract data from masks.
% NOTE: the first four letters of the name of this function must be: data_.
%
% PARAMETERS:
%   Maski_Origj           Mask of the original image.
%   Maski_Cometj          Mask of the Comet.
%   Maski_Tailj           Mask of the Tail
%   Maski_Headj           Mask of the Head.
%   Intensity_MinMax_StretchedOrig Intensity values of the stretched and
%                               original images, in order min, max.
OutValue = NaN;
```

- 5) Change in the first line of the code the name of the function from “data_TemplateFunction” to “data_FeatureName01”;

- 6) Change the last line of the code computing as “OutValue” the desired feature value (here a minimum of computer programming skills are required, but please, for any type of problem contact us, we are always open to collaborations!).

Let us see a few examples of code:

(a) For computing the average intensity value of the pixels belonging to the comet, these lines of code should be reported here:

```
Mask = Maski_Cometj;
Orig = Maski_Origj;
a = Intensity_MinMax_StretchedOrig(1);
b = Intensity_MinMax_StretchedOrig(2);
c = Intensity_MinMax_StretchedOrig(3);
d = Intensity_MinMax_StretchedOrig(4);
Orig = ((Orig-a).*((d-c)/(b-a)))+c;
if sum(Mask(:))>1 && sum(Orig(:))>1
    Inds = find(Mask>0);
    Values = Orig(Inds);
    OutValue = mean(Values);
else
    OutValue = 0;
end
```

(b) For computing the area (i.e. number of pixels) of the comet’s head, these lines of code should be reported here:

```
Mask = Maski_Headj;
if sum(Mask(:))>1
    Values = struct2cell(regionprops(Mask, 'Area'));
    OutValue = Values{1};
else
    OutValue = 0;
end
```

- 7) Finally, to automatically compute the newly implemented feature for each segmented comet, the user must run CometAnalyser using the modified source code and MATLAB, and logically not using the previously provided standalone versions.

Today, these are the features automatically computed when using the provided standalone versions of CometAnalyser:

Features extracted for the entire comet:

- *xL, xH, yL, yH*
with reference to the first pixel to the top-left corner of the image (considered the origin $[1, 1]$), coordinates of the bounding box visualised into the image around the segmented comets. For each bounding box, $[xL, yL]$ are the $[row, column]$ of the upper-left corner of the bounding box and $[xH, yH]$ are the $[row, column]$ of the lower-right-corner of the bounding box.
- *Comet_Area_inPixels*
Number of foreground pixels belonging to the comet.
- *Comet_Max_Length_inPixels*
Length in pixels of the maximum diameters through the centroid of the comet.
- *Comet_Mean_Intensity*
Average of the intensity values of the foreground pixels belonging to the comet.
- *Comet_Sphericity*
Sphericity value $[0, 1]$ computed according to the following equation:
$$Sphericity = \frac{\pi \cdot \sqrt{\frac{4 \cdot Area}{\pi}}}{Perimeter}$$
- *Comet_Std_Intensity*
Standard deviation of the intensity values of the foreground pixels belonging to the comet.
- *Comet_Sum_Intensities*
Sum of the intensity values of the foreground pixels belonging to the comet.

Features extracted for the comet's head:

- *Head_Area_inPixels*
Number of foreground pixels belonging to the comet's head.
- *Head_Max_Length_inPixels*
Length in pixels of the maximum diameters through the centroid of the comet's head.
- *Head_Mean_Intensity*
Average of the intensity values of the foreground pixels belonging to the comet's head.
- *Head_PercentDNA*

The *PercentDNA* is computed as the percentage ratio between the sum of the intensities of the comet's head, divided by the sum of the intensities of the whole comet.

$$PercentDNA = 100 \cdot \frac{Head_Sum_Intensities}{Comet_Sum_Intensities}$$

Now that for our definitions of the different masks, the sum of *Head_PercentDNA* and *Tail_PercentDNA* should not be 100%.

- *Head_Sphericity*

Sphericity value [0, 1] computed according to the following equation:

$$Sphericity = \frac{\pi \cdot \sqrt{\frac{4 \cdot Area}{\pi}}}{Perimeter}$$

- *Head_Std_Intensity*

Standard deviation of the intensity values of the foreground pixels belonging the comet's head.

- *Head_Sum_Intensities*

Sum of the intensity values of the foreground pixels belonging to the comet's head.

Features extracted for the comet's tail:

- *Tail_Area_inPixels*

Number of foreground pixels belonging to the comet's tail.

- *Tail_Extent_Moment*

The *ExtentMoment* is the ratio between the sum of the intensities of the comet's tail divided by the sum of the intensities of the whole comet, multiplied by the length in pixels of the maximum straight line inside the comet's tail. It is mathematically computed according to the following equation:

$$ExtentMoment = |Max_Line_Inside_Tail| \cdot \frac{Tail_Sum_Intensities}{Comet_Sum_Intensities}$$

- *Tail_Length_Max_Direction_Line_inPixels*

Length in pixels of the maximum straight line inside the comet's tail.

- *Tail_Mean_Intensity*

Average of the intensity values of the foreground pixels belonging to the comet's tail.

- *Tail_Olive_Moment*

The *OliveMoment* is the ratio between the sum of the intensities of the comet's tail divided by the sum of the intensities of the whole comet, multiplied per the module of the distance between the centroid of the head and the centroid of the tail. It is mathematically computed according to the following equation:

$$OliveMoment = |centroid_Head - centroid_Tail| \cdot \frac{Tail_Sum_Intensities}{Comet_Sum_Intensities}$$

- *Tail_PercentDNA*

The *PercentDNA* is computed as the percentage ratio between the sum of the intensities of the comet's tail, divided by the sum of the intensities of the whole comet.

$$PercentDNA = 100 \cdot \frac{Tail_Sum_Intensities}{Comet_Sum_Intensities}$$

Now that for our definitions of the different masks, the sum of *Head_PercentDNA* and *Tail_PercentDNA* should not be 100%.

- *Tail_Std_Intensity*

Standard deviation of the intensity values of the foreground pixels belonging to the comet's tail

- *Tail_Sum_Intensities*

Sum of the intensity values of the foreground pixels belonging to the comet's tail

13. CONTACTS

When using CometAnalyser and/or related materials, please, cite the following article:

Attila Baleon, et al., Filippo Piccinini, "CometAnalyser: a user-friendly open-source deep learning microscopy tool for quantitative comet assay analysis", 2022.

If you need further information or you have special requests, we are open to collaborations! Feel free to contact us:

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